INCH-POUND

MIL-R-23417/27A
25 April 1991

SUPERSEDING
MIL-R-23417/27
19 September 1967

MILITARY SPECIFICATION SHEET

RESOLVER, ELECTRICAL, AC
TYPE 15R26W4A

This specification is approved for use by all Departments and Agencies of the Department of Defense. The requirements for acquiring the Electrical Resolvers described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) cited in the solicitation: MIL-R-23417.

TABLE I. Requirements.

Requirements	Value	Unit	Tolerance
Excitation winding	Stator		
Rated frequency	400	Hz	±1%
Input voltage:			
Maximum	26	volts (rms)	±1%
Test	15	volts (rms)	±1%
Minimum	5	volts (rms)	±1%
Primary current	5.7	milliamps	nominal
Primary power	0.015	watts	nominal
D.C. resistance:			
Rotor	320	ohms	±15%
Stator	378	ohms	±15%
Compensator	512	ohms	±15%
Impedance:			
Zro	450+j2800	ohms	R±15%+jx±10%
Zso	470+j2545	ohms	R±15%+jx±10%
Zco	640+j2700	ohms	R±15%+jx±10%
Zrs	710+j250	ohms	nominal
Transformation ratio:			
Output/input	0.980		±1%
Rotor/compensator	0.980		±0.1%

AMSC - N/A

FSC 5990

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TABLE I. Requirements - Continued.

Requirements	Value	Unit	Tolerance
T. R. variation (R/C) due to voltage	0.05	-----	maximum
Equality of transformation ratios:			
Output/input	0.1	percent spread	maximum
Rotor/compensator	0.1	percent spread	maximum
Phase shift:			
Output/input	8.5	degrees	±0.5
Rotor/compensator	8.5	degrees	±0.5
Phase shift variation (R/C) due to voltage	±0.5	degrees	maximum
Interaxis error	±5.0	minutes	maximum
Functional error	±0.1	percent	maximum
Null voltage:			
Fundamental	8	mV(rms) output	maximum
Total	15	mV(rms) output	maximum
Shift of resolver zero:			
With voltage	±1.0	minutes	maximum
With frequency	±0.5	minutes	maximum
Harmonic distortion	0.1	percent	maximum
Temperature rise	15	degrees C	maximum
Mechanical breakaway torque:	0.15	oz-in	maximum
@ -55°C	0.30	oz-in	maximum
Radial play	0.0005	inches	maximum
End play	0.0010	inches	±0.0005
Resonant frequency	-----	kHz	minimum
Low corner frequency	58.5	Hz	maximum

NOTE:

1. A voltage value of 5 to 26 volts shall be entered in the "volts" designation space on the identification plate.

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TABLE II. Part identifying number variant characteristics.

Part Identifying No. 1/	A ±.010	B ±.010	G Maximum	Shaft Style	Terminal End	U ±.0000 -.0005
M23417/27-01A	.555	.500	2.100	Spline	Terminal Board	.1872
M23417/27-02A	.305	.250	1.850	Plain	Wire Leads	.1200
M23417/27-03A	.492	.437	1.850	Plain	Wire Leads	.1200
M23417/27-04A	.555	.500	1.850	Plain	Wire Leads	.1200
M23417/27-05A	.305	.250	2.100	Plain	Terminal Board	.1200
M23417/27-06A	.492	.437	2.100	Plain	Terminal Board	.1200
M23417/27-07A	.555	.500	2.100	Plain	Terminal Board	.1200

1/ Part identifying number M23417/27-01A shall be the NATO standard.

1/ The suffix letter following the numerical dash number corresponds to the latest modification letter in the type designation.

TABLE III. Part identifying number cross-reference. 1/

SUPERSEDED PART NUMBERS	NEW PART IDENTIFYING NUMBERS
MIL-R-23417/26	
M23417/27-01	M23417/27-01A
M23417/27-02	M23417/27-02A
M23417/27-03	M23417/27-03A
M23417/27-04	M23417/27-04A
M23417/27-05	M23417/27-05A
M23417/27-06	M23417/27-06A
M23417/27-07	M23417/27-07A

1/ All line items shown in Table III refer to equivalent and interchangeable electrical resolvers of the same type designation modification. Part number changes do not affect form, fit or function of the electrical resolvers listed therein.

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Custodians:
Army - AR
Navy - AS
Air Force - 85

Review activities:
Navy - OS, EC
Air Force - 17, 99
DLA - ES

User activities:
Army - MI, ME, AT
Navy - CG, MC
Air Force - 14, 19

Preparing activity:
Army - AR

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